

$$\text{Momentum}_S(\text{Cost}(t)) = \sum_i \text{Side}(i) \cdot (X_i - Y_i) \cdot (P_{it} - P_{i0}) \quad (9.20)$$

$$E_t(MI) = \sum_{i=1}^m \sum_{j=t}^n \frac{b_1 \cdot I_i^* \cdot x_{ij}^2}{X_i \cdot v_{ij}} + \sum_{i=1}^m \frac{Y_{it}}{Y_{it} + X_i} \cdot (1 - b_1) \cdot I_i^* \quad (9.21)$$

Notice that our market impact and timing risk equations only incorporate trading activity from the current period through the end of the trading horizon. Additionally, since permanent market impact will be reflected in market prices during trading we need to incorporate permanent impact cost in the re-optimization adjusted for quantity of shares traded. This is shown as the second expression on the right hand side in [Equation 9.21](#).

Therefore we have,

$$E_0[\text{Cost}] = C^* \quad (9.22)$$

$$E_t[\text{Cost}] = \text{Realized} + \text{Momentum}_S(\text{Cost}(t)) + E_t[MI] \quad (9.23)$$

Description of AIM and PIM for Portfolio Trading

Portfolio adaptation tactics are illustrated in [Figure 9.3](#). In this scenario, the portfolio manager is rebalancing the portfolio and also investing additional cash. This results in a trade list with an initially higher buy value than sell value. [Figure 9.3a](#) illustrates how the basket will be traded under expected market conditions. Here the buy order has an initial risk of \$150K and the sell order has an initial risk of \$100K. The manager optimizes the trade schedule using techniques described above and this results in the buys initially being transacted at a faster rate to offset the incremental risk until the residual position is hedged. Following this optimized trade schedule, the position is traded into the hedged position at 12 p.m. After this time, the buys and sells are transacted at the same trading rate.

[Figure 9.3b](#) illustrates how the basket may be traded in a situation with favorable price movement. Suppose that by 10:15 a.m. there was a decline in market prices after the open. This makes buys cheaper but sells more expensive. But since there are more shares to buy than there are to sell investors are better off. A manager employing the AIM tactic can take advantage of the better market prices and trade into the hedged position at a faster rate. Here the manager achieves the hedged position by 11:15 a.m. After this time the portfolio is traded at the more passive rate